MAE5005 Computational Fluid Dynamics, Fall 2026

Homework Set 5

Due Wednesday, April 10, 2026 by 23:59, by online submission through *Blackboard*

14. (30 points) In the lecture, we derived the following scheme for the 1D advection equation, using the three-point $(j-2, j-1, j)$ Lagrangian interpolation,

$$u_j^{n+1} = -\frac{(\Delta x - a\Delta t)a\Delta t}{2(\Delta x)^2}u_{j-2}^n + \frac{(2\Delta x - a\Delta t)a\Delta t}{(\Delta x)^2}u_{j-1}^n + \frac{(\Delta x - a\Delta t)(2\Delta x - a\Delta t)}{2(\Delta x)^2}u_j^n \tag{1}$$

   (a) First, show yourself, that this scheme can be written, alternatively, as

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{a}{2\Delta x}\left[3u_j^n - 4u_{j-1}^n + u_{j-2}^n\right] = \frac{a^2\Delta t}{2}\frac{\left[u_{j-2}^n - 2u_{j-1}^n + u_j^n\right]}{(\Delta x)^2} \tag{2}$$

   (b) Perform von Neumann stability analysis of the above scheme. In particular, using a contour plot (namely, plot the one-time-step growth factor as a function of the CFL number and $(k\Delta x)$), investigate the stability property of the above scheme.

   (c) There are two special values of $C \equiv a\Delta t/\Delta x$, for which the scheme becomes exact, namely, without any truncation error. Can you identify these two special C values? The two special values can be quickly identified by referring to the analytical solution of the original PDE.

15. (20 points) Derive the modified equation of the scheme in Problem 14, to show that the scheme is indeed of second-order accuracy both in space and in time. Namely, the modified equation should be of the form

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = A\cdot a\cdot(\Delta x)^2\frac{\partial^3 u}{\partial x^3} + B\cdot a\cdot(\Delta x)^3\frac{\partial^4 u}{\partial x^4} + \mathcal{O}[(\Delta t)^4, (\Delta t^3\Delta x), (\Delta t^2\Delta x^2), (\Delta t\Delta x^3), (\Delta x)^4], \tag{3}$$

where A and B are functions of the CFL number. Derive the explicit expressions for A and B.

16. (30 points) In the lecture, we briefly mentioned the Lax method for solving the advection equation $\partial u/\partial t + a\partial u/\partial x = 0$, namely,

$$\frac{u_j^{n+1} - \left(u_{j+1}^n + u_{j-1}^n\right)/2}{\Delta t} + a\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0. \tag{4}$$

   (a) Perform the von Neumann stability analysis to identify the stability condition for this scheme;

   (b) Perform Taylor expansions of Eq. (4) to identify the first-order error term and the second-order error term, for a fixed value of $C \equiv a\Delta t/\Delta x$. Note that this derivation is roughly 5 lines. What is the order of accuracy of the scheme in time and in space, separately?

(c) Next, derive the modified equation retaining only the *first-order* error term, namely,

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \alpha \Delta x \frac{\partial^2 u}{\partial x^2} + \mathcal{O}[(\Delta t)^2, (\Delta t \Delta x), (\Delta x)^2] \tag{5}$$

determine the expression for α. Note that this derivation is less than 5 lines. To ensure numerically stability, one would require $\alpha \geq 0$. What is then the stability condition from this perspective?